부표 2 N-kgf 환 산 표

〔표 보는 방법〕 예를들면, 10 N을 kgf로 환산할때, 제 1 BLOCK의 중앙란에 있는 10의 오른쪽 kgf란을 읽으면, 10 N은 1.0197 kgf이란 것을 알 수 있다. 또 10 kgf을 N으로 환산할 때는, 그 왼쪽의 N란을 읽으면 98.066 N인 것을 알 수 있다.

1 N=0.1019716 kgf
1 kgf=9.80665 N

N		kgf	N		kgf	N		kgf
9.8066	**1**	0.1020	333.43	**34**	3.4670	657.05	**67**	6.8321
19.613	**2**	0.2039	343.23	**35**	3.5690	666.85	**68**	6.9341
29.420	**3**	0.3059	353.04	**36**	3.6710	676.66	**69**	7.0360
39.227	**4**	0.4079	362.85	**37**	3.7729	686.47	**70**	7.1380
49.033	**5**	0.5099	372.65	**38**	3.8749	696.27	**71**	7.2400
58.840	**6**	0.6118	382.46	**39**	3.9769	706.08	**72**	7.3420
68.647	**7**	0.7138	392.27	**40**	4.0789	715.89	**73**	7.4439
78.453	**8**	0.8158	402.07	**41**	4.1808	725.69	**74**	7.5459
88.260	**9**	0.9177	411.88	**42**	4.2828	735.50	**75**	7.6479
98.066	**10**	1.0197	421.69	**43**	4.3848	745.31	**76**	7.7498
107.87	**11**	1.1217	431.49	**44**	4.4868	755.11	**77**	7.8518
117.68	**12**	1.2237	441.30	**45**	4.5887	764.92	**78**	7.9538
127.49	**13**	1.3256	451.11	**46**	4.6907	774.73	**79**	8.0558
137.29	**14**	1.4276	460.91	**47**	4.7927	784.53	**80**	8.1577
147.10	**15**	1.5296	470.72	**48**	4.8946	794.34	**81**	8.2597
156.91	**16**	1.6315	480.53	**49**	4.9966	804.15	**82**	8.3617
166.71	**17**	1.7335	490.33	**50**	5.0986	813.95	**83**	8.4636
176.52	**18**	1.8355	500.14	**51**	5.2006	823.76	**84**	8.5656
186.33	**19**	1.9375	509.95	**52**	5.3025	833.57	**85**	8.6676
196.13	**20**	2.0394	519.75	**53**	5.4045	843.37	**86**	8.7696
205.94	**21**	2.1414	529.56	**54**	5.5065	853.18	**87**	8.8715
215.75	**22**	2.2434	539.37	**55**	5.6084	862.99	**88**	8.9735
225.55	**23**	2.3453	549.17	**56**	5.7104	872.79	**89**	9.0755
235.36	**24**	2.4473	558.98	**57**	5.8124	882.60	**90**	9.1774
245.17	**25**	2.5493	568.79	**58**	5.9144	892.41	**91**	9.2794
254.97	**26**	2.6513	578.59	**59**	6.0163	902.21	**92**	9.3814
264.78	**27**	2.7532	588.40	**60**	6.1183	912.02	**93**	9.4834
274.59	**28**	2.8552	598.21	**61**	6.2203	921.83	**94**	9.5853
284.39	**29**	2.9572	608.01	**62**	6.3222	931.63	**95**	9.6873
294.20	**30**	3.0591	617.82	**63**	6.4242	941.44	**96**	9.7893
304.01	**31**	3.1611	627.63	**64**	6.5262	951.25	**97**	9.8912
313.81	**32**	3.2631	637.43	**65**	6.6282	961.05	**98**	9.9932
323.62	**33**	3.3651	647.24	**66**	6.7301	970.86	**99**	10.095

부표 3 kg-lb 환 산 표

〔표 보는 방법〕 예를들면, kg를 lb로 환산할때, 제 1 BLOCK의 중앙란에 있는 10 의 오른쪽 lb란을 읽으면, 10 kg는 22.0461 lb인 것을 알 수 있다. 또 10 lb를 kg으로 환산할 때는, 그 왼쪽의 kg란을 읽으면 10 lb는 4.536 kg인 것을 알 수 있다.

1 kg=2.2046226 lb
1 lb=0.45359237 kg

kg		lb	kg		lb	kg		lb
0.454	**1**	2.205	15.422	**34**	74.957	30.391	**67**	147.71
0.907	**2**	4.409	15.876	**35**	77.162	30.844	**68**	149.91
1.361	**3**	6.614	16.329	**36**	79.366	31.298	**69**	152.12
1.814	**4**	8.818	16.783	**37**	81.571	31.751	**70**	154.32
2.268	**5**	11.023	17.237	**38**	83.776	32.205	**71**	156.53
2.722	**6**	13.228	17.690	**39**	85.980	32.659	**72**	158.73
3.175	**7**	15.432	18.144	**40**	88.185	33.112	**73**	160.94
3.629	**8**	17.637	18.597	**41**	90.390	33.566	**74**	163.14
4.082	**9**	19.842	19.051	**42**	92.594	34.019	**75**	165.35
4.536	**10**	22.046	19.504	**43**	94.799	34.473	**76**	167.55
4.990	**11**	24.251	19.958	**44**	97.003	34.927	**77**	169.76
5.443	**12**	26.455	20.412	**45**	99.208	35.380	**78**	171.96
5.897	**13**	28.660	20.865	**46**	101.41	35.834	**79**	174.17
6.350	**14**	30.865	21.319	**47**	103.62	36.287	**80**	176.37
6.804	**15**	33.069	21.772	**48**	105.82	36.741	**81**	178.57
7.257	**16**	35.274	22.226	**49**	108.03	37.195	**82**	180.78
7.711	**17**	37.479	22.680	**50**	110.23	37.648	**83**	182.98
8.165	**18**	39.683	23.133	**51**	112.44	38.102	**84**	185.19
8.618	**19**	41.888	23.587	**52**	114.64	38.555	**85**	187.39
9.072	**20**	44.092	24.040	**53**	116.84	39.009	**86**	189.60
9.525	**21**	46.297	24.494	**54**	119.05	39.463	**87**	191.80
9.979	**22**	48.502	24.948	**55**	121.25	39.916	**88**	194.01
10.433	**23**	50.706	25.401	**56**	123.46	40.370	**89**	196.21
10.886	**24**	52.911	25.855	**57**	125.66	40.823	**90**	198.42
11.340	**25**	55.116	26.308	**58**	127.87	41.277	**91**	200.62
11.793	**26**	57.320	26.762	**59**	130.07	41.730	**92**	202.83
12.247	**27**	59.525	27.216	**60**	132.28	42.184	**93**	205.03
12.701	**28**	61.729	27.669	**61**	134.48	42.638	**94**	207.23
13.154	**29**	63.934	28.123	**62**	136.69	43.091	**95**	209.44
13.608	**30**	66.139	28.576	**63**	138.89	43.545	**96**	211.64
14.061	**31**	68.343	29.030	**64**	141.10	43.998	**97**	213.85
14.515	**32**	70.548	29.484	**65**	143.30	44.452	**98**	216.05
14.969	**33**	72.753	29.937	**66**	145.51	44.906	**99**	218.26

부표 4 ℃~℉ 온 도 환 산 표

〔표 보는 방법〕 예를들면, 38℃을 ℉로 환산할때, 제 2 BLOCK의 중앙란에 있는 38의 오른쪽 ℉란을 읽으면, 38℃은 100.4℉이란 것을 알 수 있다.
또 38℉을 ℃으로 환산할 때는, 그 왼쪽의 ℃란을 읽으면 38℉는 3.3℃인 것을 알 수 있다.

$$C=\frac{5}{9}(F-32)$$

$$F=32+\frac{9}{5}C$$

℃		℉	℃		℉	℃		℉	℃		℉
−73.3	**−100**	−148.0	0.0	**32**	89.6	21.7	**71**	159.8	43.3	**110**	230
−62.2	**− 80**	−112.0	0.6	**33**	91.4	22.2	**72**	161.6	46.1	**115**	239
−51.1	**− 60**	− 76.0	1.1	**34**	93.2	22.8	**73**	163.4	48.9	**120**	248
−40.0	**− 40**	− 40.0	1.7	**35**	95.0	23.3	**74**	165.2	51.7	**125**	257
−34.4	**− 30**	− 22.0	2.2	**36**	96.8	23.9	**75**	167.0	54.4	**130**	266
−28.9	**− 20**	− 4.0	2.8	**37**	98.6	24.4	**76**	168.8	57.2	**135**	275
−23.3	**− 10**	14.0	3.3	**38**	100.4	25.0	**77**	170.6	60.0	**140**	284
−17.8	**0**	32.0	3.9	**39**	102.2	25.6	**78**	172.4	65.6	**150**	302
−17.2	**1**	33.8	4.4	**40**	104.0	26.1	**79**	174.2	71.1	**160**	320
−16.7	**2**	35.6	5.0	**41**	105.8	26.7	**80**	176.0	76.7	**170**	338
−16.1	**3**	37.4	5.6	**42**	107.6	27.2	**81**	177.8	82.2	**180**	356
−15.6	**4**	39.2	6.1	**43**	109.4	27.8	**82**	179.6	87.8	**190**	374
−15.0	**5**	41.0	6.7	**44**	111.2	28.3	**83**	181.4	93.3	**200**	392
−14.4	**6**	42.8	7.2	**45**	113.0	28.9	**84**	183.2	98.9	**210**	410
−13.9	**7**	44.6	7.8	**46**	114.8	29.4	**85**	185.0	104.4	**220**	428
−13.3	**8**	46.4	8.3	**47**	116.6	30.0	**86**	186.8	110.0	**230**	446
−12.8	**9**	48.2	8.9	**48**	118.4	30.6	**87**	188.6	115.6	**240**	464
−12.2	**10**	50.0	9.4	**49**	120.2	31.1	**88**	190.4	121.1	**250**	482
−11.7	**11**	51.8	10.0	**50**	122.0	31.7	**89**	192.2	148.9	**300**	572
−11.1	**12**	53.6	10.6	**51**	123.8	32.2	**90**	194.0	176.7	**350**	662
−10.6	**13**	55.4	11.1	**52**	125.6	32.8	**91**	195.8	204	**400**	752
−10.0	**14**	57.2	11.7	**53**	127.4	33.3	**92**	197.6	232	**450**	842
− 9.4	**15**	59.0	12.2	**54**	129.2	33.9	**93**	199.4	260	**500**	932
− 8.9	**16**	60.8	12.8	**55**	131.0	34.4	**94**	201.2	288	**550**	1022
− 8.3	**17**	62.6	13.3	**56**	132.8	35.0	**95**	203.0	316	**600**	1112
− 7.8	**18**	64.4	13.9	**57**	134.6	35.6	**96**	204.8	343	**650**	1202
− 7.2	**19**	66.2	14.4	**58**	136.4	36.1	**97**	206.6	371	**700**	1292
− 6.7	**20**	68.0	15.0	**59**	138.2	36.7	**98**	208.4	399	**750**	1382
− 6.1	**21**	69.8	15.6	**60**	140.0	37.2	**99**	210.2	427	**800**	1472
− 5.6	**22**	71.6	16.1	**61**	141.8	37.8	**100**	212.0	454	**850**	1562
− 5.0	**23**	73.4	16.7	**62**	143.6	38.3	**101**	213.8	482	**900**	1652
− 4.4	**24**	75.2	17.2	**63**	145.4	38.9	**102**	215.6	510	**950**	1742
− 3.9	**25**	77.0	17.8	**64**	147.2	39.4	**103**	217.4	538	**1000**	1832
− 3.3	**26**	78.8	18.3	**65**	149.0	40.0	**104**	219.2	593	**1100**	2012
− 2.8	**27**	80.6	18.9	**66**	150.8	40.6	**105**	221.0	649	**1200**	2192
− 2.2	**28**	82.4	19.4	**67**	152.6	41.1	**106**	222.8	704	**1300**	2372
− 1.7	**29**	84.2	20.0	**68**	154.4	41.7	**107**	224.6	760	**1400**	2552
− 1.1	**30**	86.0	20.6	**69**	156.2	42.2	**108**	226.4	816	**1500**	2732
− 0.6	**31**	87.8	21.1	**70**	158.0	42.8	**109**	228.2	871	**1600**	2912

부표 5 점 도 환 산 표

동점도 mm^2/s	세이볼트 유니버셜 SUS (초)		1 호 형 레이우드 R (초)		앵귤러 E (도)
	100°F	210°F	50°C	100°C	
2	32.6	32.8	30.8	31.2	1.14
3	36.0	36.3	33.3	33.7	1.22
4	39.1	39.4	35.9	36.5	1.31
5	42.3	42.6	38.5	39.1	1.40
6	45.5	45.8	41.1	41.7	1.48
7	48.7	49.0	43.7	44.3	1.56
8	52.0	52.4	46.3	47.0	1.65
9	55.4	55.8	49.1	50.0	1.75
10	58.8	59.2	52.1	52.9	1.84
11	62.3	62.7	55.1	56.0	1.93
12	65.9	66.4	58.2	59.1	2.02
13	69.6	70.1	61.4	62.3	2.12
14	73.4	73.9	64.7	65.6	2.22
15	77.2	77.7	68.0	69.1	2.32
16	81.1	81.7	71.5	72.6	2.43
17	85.1	85.7	75.0	76.1	2.54
18	89.2	89.8	78.6	79.7	2.64
19	93.3	94.0	82.1	83.6	2.76
20	97.5	98.2	85.8	87.4	2.87
21	102	102	89.5	91.3	2.98
22	106	107	93.3	95.1	3.10
23	110	111	97.1	98.9	3.22
24	115	115	101	103	3.34
25	119	120	105	107	3.46
26	123	124	109	111	3.58
27	128	129	112	115	3.70
28	132	133	116	119	3.82
29	137	138	120	123	3.95
30	141	142	124	127	4.07
31	145	146	128	131	4.20
32	150	150	132	135	4.32
33	154	155	136	139	4.45
34	159	160	140	143	4.57

동점도 mm^2/s	세이볼트 유니버셜 SUS (초)		1 호 형 레이우드 R (초)		앵귤러 E (도)
	100°F	210°F	50°C	100°C	
35	163	164	144	147	4.70
36	168	170	148	151	4.83
37	172	173	153	155	4.96
38	177	178	156	159	5.08
39	181	183	160	164	5.21
40	186	187	164	168	5.34
41	190	192	168	172	5.47
42	195	196	172	176	5.59
43	199	201	176	180	5.72
44	204	205	180	185	5.85
45	208	210	184	189	5.98
46	213	215	188	193	6.11
47	218	219	193	197	6.24
48	222	224	197	202	6.37
49	227	228	201	206	6.50
50	231	233	205	210	6.63
55	254	256	225	231	7.24
60	277	279	245	252	7.90
65	300	302	266	273	8.55
70	323	326	286	294	9.21
75	346	349	306	315	9.89
80	371	373	326	336	10.5
85	394	397	347	357	11.2
90	417	420	367	378	11.8
95	440	443	387	399	12.5
100	464	467	408	420	13.2
120	556	560	490	504	15.8
140	649	653	571	588	18.4
160	742	747	653	672	21.1
180	834	840	734	757	23.7
200	927	933	816	841	26.3
250	1 159	1 167	1 020	1 051	32.9
300	1 391	1 400	1 224	1 241	39.5

비 고 $1mm^2/s=1cSt$

부표 6 inch-mm 환 산 표

1″ = 25.4mm

inch		0	1	2	3	4	5	6	7	8	9	10
분수	소수	mm										
0	**0.000000**	**0.000**	**25.400**	**50.800**	**76.200**	**101.600**	**127.000**	**152.400**	**177.800**	**203.200**	**228.600**	**254.000**
1/64	0.015625	0.397	25.797	51.197	76.597	101.997	127.397	152.797	178.197	203.597	228.997	254.397
1/32	0.031250	0.794	26.194	51.594	76.994	102.394	127.794	153.194	178.594	203.994	229.394	254.794
3/64	0.046875	1.191	26.591	51.991	77.391	102.791	128.191	153.591	178.991	204.391	229.791	255.191
1/16	**0.062500**	**1.588**	**26.988**	**52.388**	**77.788**	**103.188**	**128.588**	**153.988**	**179.388**	**204.788**	**230.188**	**255.588**
5/64	0.078125	1.984	27.384	52.784	78.184	103.584	128.984	154.384	179.784	205.184	230.584	255.984
3/32	0.093750	2.381	27.781	53.181	78.581	103.981	129.381	154.781	180.181	205.581	230.981	256.381
7/64	0.109375	2.778	28.178	53.578	78.978	104.378	129.778	155.178	180.578	205.978	231.378	256.778
1/8	**0.125000**	**3.175**	**28.575**	**53.975**	**79.375**	**104.775**	**130.175**	**155.575**	**180.975**	**206.375**	**231.775**	**257.175**
9/64	0.140625	3.572	28.972	54.372	79.772	105.172	130.572	155.972	181.372	206.772	232.172	257.572
5/32	0.156250	3.969	29.369	54.769	80.169	105.569	130.969	156.369	181.769	207.169	232.569	257.969
11/64	0.171875	4.366	29.766	55.166	80.566	105.966	131.366	156.766	182.166	207.566	232.966	258.366
3/16	**0.187500**	**4.762**	**30.162**	**55.562**	**80.962**	**106.362**	**131.762**	**157.162**	**182.562**	**207.962**	**233.362**	**258.762**
13/64	0.203125	5.159	30.559	55.959	81.359	106.759	132.159	157.559	182.959	208.359	233.759	259.159
7/32	0.218750	5.556	30.956	56.356	81.756	107.156	132.556	157.956	183.356	208.756	234.156	259.556
15/64	0.234375	5.953	31.353	56.753	82.153	107.553	132.953	158.353	183.753	209.153	234.553	259.953
1/4	**0.250000**	**6.350**	**31.750**	**57.150**	**82.550**	**107.950**	**133.350**	**158.750**	**184.150**	**209.550**	**234.950**	**260.350**
17/64	0.265625	6.747	32.147	57.547	82.947	108.347	133.747	159.147	184.547	209.947	235.347	260.747
9/32	0.281250	7.144	32.544	57.944	83.344	108.744	134.144	159.544	184.944	210.344	235.744	261.144
19/64	0.296875	7.541	32.941	58.341	83.741	109.141	134.541	159.941	185.341	210.741	236.141	261.541
5/16	**0.312500**	**7.938**	**33.338**	**58.738**	**84.138**	**109.538**	**134.938**	**160.338**	**185.738**	**211.138**	**236.538**	**261.938**
21/64	0.328125	8.334	33.734	59.134	84.534	109.934	135.334	160.734	186.134	211.534	236.934	262.334
11/32	0.343750	8.731	34.131	59.531	84.931	110.331	135.731	161.131	186.531	211.931	237.331	262.731
23/64	0.359375	9.128	34.528	59.928	85.328	110.728	136.128	161.528	186.928	212.328	237.728	263.128
3/8	**0.375000**	**9.525**	**34.925**	**60.325**	**85.725**	**111.125**	**136.525**	**161.925**	**187.325**	**212.725**	**238.125**	**263.525**
25/64	0.390625	9.922	35.322	60.722	86.122	111.522	136.922	162.322	187.722	213.122	238.522	263.922
13/32	0.406250	10.319	35.719	61.119	86.519	111.919	137.319	162.719	188.119	213.519	238.919	264.319
27/64	0.421875	10.716	36.116	61.516	86.916	112.316	137.716	163.116	188.516	213.916	239.316	264.716
7/16	**0.437500**	**11.112**	**36.512**	**61.912**	**87.312**	**112.712**	**138.112**	**163.512**	**188.912**	**214.312**	**239.712**	**265.112**
29/64	0.453125	11.509	36.909	62.309	87.709	113.109	138.509	163.909	189.309	214.709	240.109	265.509
15/32	0.468750	11.906	37.306	62.706	88.106	113.506	138.906	164.306	189.706	215.106	240.506	265.906
31/64	0.484375	12.303	37.703	63.103	88.503	113.903	139.303	164.703	190.103	215.503	240.903	266.303
1/2	**0.500000**	**12.700**	**38.100**	**63.500**	**88.900**	**114.300**	**139.700**	**165.100**	**190.500**	**215.900**	**241.300**	**266.700**
33/64	0.515625	13.097	38.497	63.897	89.297	114.697	140.097	165.497	190.897	216.297	241.697	267.097
17/32	0.531250	13.494	38.894	64.294	89.694	115.094	140.494	165.894	191.294	216.694	242.094	267.494
35/64	0.546875	13.891	39.291	64.691	90.091	115.491	140.891	166.291	191.691	217.091	242.491	267.891
9/16	**0.562500**	**14.288**	**39.688**	**65.088**	**90.488**	**115.888**	**141.288**	**166.688**	**192.088**	**217.488**	**242.888**	**268.288**
37/64	0.578125	14.684	40.084	65.484	90.884	116.284	141.684	167.084	192.484	217.884	243.284	268.684
19/32	0.593750	15.081	40.481	65.881	91.281	116.681	142.081	167.481	192.881	218.281	243.681	269.081
39/64	0.609375	15.478	40.878	66.278	91.678	117.078	142.478	167.878	193.278	218.678	244.078	269.478
5/8	**0.625000**	**15.875**	**41.275**	**66.675**	**92.075**	**117.475**	**142.875**	**168.275**	**193.675**	**219.075**	**244.475**	**269.875**
41/64	0.640625	16.272	41.672	67.072	92.472	117.872	143.272	168.672	194.072	219.472	244.872	270.272
21/32	0.656250	16.669	42.069	67.469	92.869	118.269	143.669	169.069	194.469	219.869	245.269	270.669
43/64	0.671875	17.066	42.466	67.866	93.266	118.666	144.066	169.466	194.866	220.266	245.666	271.066
11/16	**0.687500**	**17.462**	**42.862**	**68.262**	**93.662**	**119.062**	**144.462**	**169.862**	**195.262**	**220.662**	**246.062**	**271.462**
45/64	0.703125	17.859	43.259	68.659	94.059	119.459	144.859	170.259	195.659	221.059	246.459	271.859
23/32	0.718750	18.256	43.656	69.056	94.456	119.856	145.256	170.656	196.056	221.456	246.856	272.256
47/64	0.734375	18.653	44.053	69.453	94.853	120.253	145.653	171.053	196.453	221.853	247.253	272.653
3/4	**0.750000**	**19.050**	**44.450**	**69.850**	**95.250**	**120.650**	**146.050**	**171.450**	**196.850**	**222.250**	**247.650**	**273.050**
49/64	0.765625	19.447	44.847	70.247	95.647	121.047	146.447	171.847	197.247	222.647	248.047	273.447
25/32	0.781250	19.844	45.244	70.644	96.044	121.444	146.844	172.244	197.644	223.044	248.444	273.844
51/64	0.796875	20.241	45.641	71.041	96.441	121.841	147.241	172.641	198.041	223.441	248.841	274.241
13/16	**0.812500**	**20.638**	**46.038**	**71.438**	**96.838**	**122.238**	**147.638**	**173.038**	**198.438**	**223.838**	**249.238**	**274.638**
53/64	0.828125	21.034	46.434	71.834	97.234	122.634	148.034	173.434	198.834	224.234	249.634	275.034
27/32	0.843750	21.431	46.831	72.231	97.631	123.031	148.431	173.831	199.231	224.631	250.031	275.431
55/64	0.859375	21.828	47.228	72.628	98.028	123.428	148.828	174.228	199.628	225.028	250.428	275.828
7/8	**0.875000**	**22.225**	**47.625**	**73.025**	**98.425**	**123.825**	**149.225**	**174.625**	**200.025**	**225.425**	**250.825**	**276.225**
57/64	0.890625	22.622	48.022	73.422	98.822	124.222	149.622	175.022	200.422	225.822	251.222	276.622
29/32	0.906250	23.019	48.419	73.819	99.219	124.619	150.019	175.419	200.819	226.219	251.619	277.019
59/64	0.921875	23.416	48.816	74.216	99.616	125.016	150.416	175.816	201.216	226.616	252.016	277.416
15/16	**0.937500**	**23.812**	**49.212**	**74.612**	**100.012**	**125.412**	**150.812**	**176.212**	**201.612**	**227.012**	**252.412**	**277.812**
61/64	0.953125	24.209	49.609	75.009	100.409	125.809	151.209	176.609	202.009	227.409	252.809	278.209
31/32	0.968750	24.606	50.006	75.406	100.806	126.206	151.606	177.006	202.406	227.806	253.206	278.606
63/64	0.984375	25.003	50.403	75.803	101.203	126.603	152.003	177.403	202.803	228.203	253.603	279.003

1″ = 25.4mm

inch		11	12	13	14	15	16	17	18	19	20
분수	소수	mm									
0	**0.0000**	**279.400**	**304.800**	**330.200**	**355.600**	**381.000**	**406.400**	**431.800**	**457.200**	**482.600**	**508.000**
1/16	0.0625	280.988	306.388	331.788	357.188	382.588	407.988	433.388	458.788	484.188	509.588
1/8	0.1250	282.575	307.975	333.375	358.775	384.175	409.575	434.975	460.375	485.775	511.175
3/16	0.1875	284.162	309.562	334.962	360.362	385.762	411.162	436.562	461.962	487.362	512.762
1/4	**0.2500**	**285.750**	**311.150**	**336.550**	**361.950**	**387.350**	**412.750**	**438.150**	**463.550**	**488.950**	**514.350**
5/16	0.3125	287.338	312.738	338.138	363.538	388.938	414.338	439.738	465.138	490.538	515.938
3/8	0.3750	288.925	314.325	339.725	365.125	390.525	415.925	441.325	466.725	492.125	517.525
7/16	0.4375	290.512	315.912	341.312	366.712	392.112	417.512	442.912	468.312	493.712	519.112
1/2	**0.5000**	**292.100**	**317.500**	**342.900**	**368.300**	**393.700**	**419.100**	**444.500**	**469.900**	**495.300**	**520.700**
9/16	0.5625	293.688	319.088	344.488	369.888	395.288	420.688	446.088	471.488	496.888	522.288
5/8	0.6250	295.275	320.675	346.075	371.475	396.875	422.275	447.675	473.075	498.475	523.875
11/16	0.6875	296.862	322.262	347.662	373.062	398.462	423.862	449.262	474.662	500.062	525.462
3/4	**0.7500**	**298.450**	**323.850**	**349.250**	**374.650**	**400.050**	**425.450**	**450.850**	**476.250**	**501.650**	**527.050**
13/16	0.8125	300.038	325.438	350.838	376.238	401.638	427.038	452.438	477.838	503.238	528.638
7/8	0.8750	301.625	327.025	352.425	377.825	403.225	428.625	454.025	479.425	504.825	530.225
15/16	0.9375	303.212	328.612	354.012	379.412	404.812	430.212	455.612	481.012	506.412	531.812

1″ = 25.4mm

inch		21	22	23	24	25	26	27	28	29	30
분수	소수	mm									
0	**0.0000**	**533.400**	**558.800**	**584.200**	**609.600**	**635.000**	**660.400**	**685.800**	**711.200**	**736.600**	**762.000**
1/16	0.0625	534.988	560.388	585.788	611.188	636.588	661.988	687.388	712.788	738.188	763.588
1/8	0.1250	536.575	561.975	587.375	612.775	638.175	663.575	688.975	714.375	739.775	765.175
3/16	0.1875	538.162	563.562	588.962	614.362	639.762	665.162	690.562	715.962	741.362	766.762
1/4	**0.2500**	**539.750**	**565.150**	**590.550**	**615.950**	**641.350**	**666.750**	**692.150**	**717.550**	**742.950**	**768.350**
5/16	0.3125	541.338	566.738	592.138	617.538	642.938	668.338	693.738	719.138	744.538	769.938
3/8	0.3750	542.925	568.325	593.725	619.125	644.525	669.925	695.325	720.725	746.125	771.525
7/16	0.4375	544.512	569.912	595.312	620.712	646.112	671.512	696.912	722.312	747.712	773.112
1/2	**0.5000**	**546.100**	**571.500**	**596.900**	**622.300**	**647.700**	**673.100**	**698.500**	**723.900**	**749.300**	**774.700**
9/16	0.5625	547.688	573.088	598.488	623.888	649.288	674.688	700.088	725.488	750.888	776.288
5/8	0.6250	549.275	574.675	600.075	625.475	650.875	676.275	701.675	727.075	752.475	777.875
11/16	0.6875	550.862	576.262	601.662	627.062	652.462	677.862	703.262	728.662	754.062	779.462
3/4	**0.7500**	**552.450**	**577.850**	**603.250**	**628.650**	**654.050**	**679.450**	**704.850**	**730.250**	**755.650**	**781.050**
13/16	0.8125	554.038	579.438	604.838	630.238	655.638	681.038	706.438	731.838	757.238	782.638
7/8	0.8750	555.625	581.025	606.425	631.825	657.225	682.625	708.025	733.425	758.825	784.225
15/16	0.9375	557.212	582.612	608.012	633.412	658.812	684.212	709.612	735.012	760.412	785.812

1″ = 25.4mm

inch		31	32	33	34	35	36	37	38	39	40
분수	소수	mm									
0	**0.0000**	**787.400**	**812.800**	**838.200**	**863.600**	**889.000**	**914.400**	**939.800**	**965.200**	**990.600**	**1016.000**
1/16	0.0625	788.988	814.388	839.788	865.188	890.588	915.988	941.388	966.788	992.188	1017.588
1/8	0.1250	790.575	815.975	841.375	866.775	892.175	917.575	942.975	968.375	993.775	1019.175
3/16	0.1875	792.162	817.562	842.962	868.362	893.762	919.162	944.562	969.962	995.362	1020.762
1/4	**0.2500**	**793.750**	**819.150**	**844.550**	**869.950**	**895.350**	**920.750**	**946.150**	**971.550**	**996.950**	**1022.350**
5/16	0.3125	795.338	820.738	846.138	871.538	896.938	922.338	947.738	973.138	998.538	1023.938
3/8	0.3750	796.925	822.325	847.725	873.125	898.525	923.925	949.325	974.725	1000.125	1025.525
7/16	0.4375	798.512	823.912	849.312	874.712	900.112	925.512	950.912	976.312	1001.712	1027.112
1/2	**0.5000**	**800.100**	**825.500**	**850.900**	**876.300**	**901.700**	**927.100**	**952.500**	**977.900**	**1003.300**	**1028.700**
9/16	0.5625	801.688	827.088	852.488	877.888	903.288	928.688	954.088	979.488	1004.888	1030.288
5/8	0.6250	803.275	828.675	854.075	879.475	904.875	930.275	955.675	981.075	1006.475	1031.875
11/16	0.6875	804.862	830.262	855.662	881.062	906.462	931.862	957.262	982.662	1008.062	1033.462
3/4	**0.7500**	**806.450**	**831.850**	**857.250**	**882.650**	**908.050**	**933.450**	**958.850**	**984.250**	**1009.650**	**1035.050**
13/16	0.8125	808.038	833.438	858.838	884.238	909.638	935.038	960.438	985.838	1011.238	1036.638
7/8	0.8750	809.625	835.025	860.425	885.825	911.225	936.625	962.025	987.425	1012.825	1038.225
15/16	0.9375	811.212	836.612	862.012	887.412	912.812	938.212	963.621	989.012	1014.412	1039.812

부 표 7 경 도 환 산 표 (참고)

록크웰 C 스케일경도 (1 471N) {150kgf}	빅 커 스 경 도	브리넬경도		록크웰경도		쇼어경도
		표 준 구	텅 스 텐 카바이트구	A스케일 하중588.4N {60kgf} brale압예	B스케일 하중980.7N {100kgf} 경1.588mm (1/16in)구	
68	940	—	—	85.6	—	97
67	900	—	—	85.0	—	95
66	865	—	—	84.5	—	92
65	832	—	739	83.9	—	91
64	800	—	722	83.4	—	88
63	772	—	705	82.8	—	87
62	746	—	688	82.3	—	85
61	720	—	670	81.8	—	83
60	697	—	654	81.2	—	81
59	674	—	634	80.7	—	80
58	653	—	615	80.1	—	78
57	633	—	595	79.6	—	76
56	613	—	577	79.0	—	75
55	595	—	560	78.5	—	74
54	577	—	543	78.0	—	72
53	560	—	525	77.4	—	71
52	544	500	512	76.8	—	69
51	528	487	496	76.3	—	68
50	513	475	481	75.9	—	67
49	498	464	469	75.2	—	66
48	484	451	455	74.7	—	64
47	471	442	443	74.1	—	63
46	458	432	432	73.6	—	62
45	446	421	421	73.1	—	60
44	434	409	409	72.5	—	58
43	423	400	400	72.0	—	57
42	412	390	390	71.5	—	56
41	402	381	381	70.9	—	55
40	392	371	371	70.4	—	54
39	382	362	362	69.9	—	52
38	372	353	353	69.4	—	51
37	363	344	344	68.9	—	50
36	354	336	336	68.4	(109.0)	49
35	345	327	327	67.9	(108.5)	48
34	336	319	319	67.4	(108.0)	47
33	327	311	311	66.8	(107.5)	46
32	318	301	301	66.3	(107.0)	44
31	310	294	294	65.8	(106.0)	43
30	302	286	286	65.3	(105.5)	42
29	294	279	279	64.7	(104.5)	41
28	286	271	271	64.3	(104.0)	41
27	279	264	264	63.8	(103.0)	40
26	272	258	258	63.3	(102.5)	38
25	266	253	253	62.8	(101.5)	38
24	260	247	247	62.4	(101.0)	37
23	254	243	243	62.0	100.0	36
22	248	237	237	61.5	99.0	35
21	243	231	231	61.0	98.5	35
20	238	226	226	60.5	97.8	34
(**18**)	230	219	219	—	96.7	33
(**16**)	222	212	212	—	95.5	32
(**14**)	213	203	203	—	93.9	31
(**12**)	204	194	194	—	92.3	29
(**10**)	196	187	187	—	90.7	28
(**8**)	188	179	179	—	89.5	27
(**6**)	180	171	171	—	87.1	26
(**4**)	173	165	165	—	85.5	25
(**2**)	166	158	158	—	83.5	24
(**0**)	160	152	152	—	81.7	24

부표8 금속재료의 물리적 기계적 성질

재 료	비 중	선팽창계수 (0°~100°C) (K^{-1})	硬 度 (브리넬)	종탄성계수 (MPa) {kgf/mm²}	인장강도 (MPa) {kgf/mm²}	항 복 점 (MPa) {kgf/mm²}	팽창율 (%)
베 어 링 강 (燒 入)	7.83	12.5×10^{-6}	650~740	208 000 {21 200}	1 570~1 960 {160~200}	—	—
마 르 텐 사 이 트 계 스 테 인 레 스 동 SUS 440C	7.68	10.1×10^{-6}	580	200 000 {20 400}	1 960 {200}	1 860 {190}	—
軟 鋼 (C=0.12~0.20%)	7.86	11.6×10^{-6}	100~130	206 000 {21 000}	373~471 {38~48}	216~294 {22~30}	24~36
硬 鋼 (C=0.3~0.5%)	7.84	11.3×10^{-6}	160~200	206 000 {21 000}	539~686 {55~70}	333~451 {34~46}	14~26
오 스 테 나 이 트 계 스 테 인 레 스 동 SUS 304	8.03	16.3×10^{-6}	150	193 000 {19 700}	588 {60}	245 {25}	60
주 철 — 회 주 철 FC200	7.3	10.4×10^{-6}	223	98 100 {10 000}	200이상 {20}	—	—
주 철 — 구상흑연주철 FCD400	7.0	11.7×10^{-6}	201이하		400이상 {41}	—	12이상
알 루 미 늄	2.69	23.7×10^{-6}	15~26	70 600 {7 200}	78 {8}	34 {3.5}	35
아 연	7.14	31×10^{-6}	30~60	92 200 {9 400}	147 {15}	—	30~40
동	8.93	16.2×10^{-6}	50	123 000 {12 500}	196 {20}	69 {7}	15~20
황 동 (燒 鈍)	8.5	19.1×10^{-6}	약45	103 000 {10 500}	294~343 {30~35}	—	65~75
황 동 (加 工)			85~130		363~539 {37~55}		15~50

비 고 Quenching된 베어링강 및 마르텐사이트계 스테인레스동의 경도는, 록크웰 C스케일 경도로 표시되는 것이 보통이지만 비교해 보기 위하여 브리넬 경도로 환산하여 표시하였다.

부표 9 축 의

경의 구분(mm) 초과	이하	베어링의평면내평균내경의치수차(0급) Δ_{dmp}	d6	e6	f6	g5	g6	h5	h6	h7	h8	h9	h10	js5	js6
3	6	0 −8	−30 −38	−20 −28	−10 −18	−4 −9	−4 −12	0 −5	0 −8	0 −12	0 −18	0 −30	0 −48	±2.5	±4
6	10	0 −8	−40 −49	−25 −34	−13 −22	−5 −11	−5 −14	0 −6	0 −9	0 −15	0 −22	0 −36	0 −58	±3	±4.5
10	18	0 −8	−50 −61	−32 −43	−16 −27	−6 −14	−6 −17	0 −8	0 −11	0 −18	0 −27	0 −43	0 −70	±4	±5.5
18	30	0 −10	−65 −78	−40 −53	−20 −33	−7 −16	−7 −20	0 −9	0 −13	0 −21	0 −33	0 −52	0 −84	±4.5	±6.5
30	50	0 −12	−80 −96	−50 −66	−25 −41	−9 −20	−9 −25	0 −11	0 −16	0 −25	0 −39	0 −62	0 −100	±5.5	±8
50	80	0 −15	−100 −119	−60 −79	−30 −49	−10 −23	−10 −29	0 −13	0 −19	0 −30	0 −46	0 −74	0 −120	±6.5	±9.5
80	120	0 −20	−120 −142	−72 −94	−36 −58	−12 −27	−12 −34	0 −15	0 −22	0 −35	0 −54	0 −87	0 −140	±7.5	±11
120	180	0 −25	−145 −170	−85 −110	−43 −68	−14 −32	−14 −39	0 −18	0 −25	0 −40	0 −63	0 −100	0 −160	±9	±12.5
180	250	0 −30	−170 −199	−100 −129	−50 −79	−15 −35	−15 −44	0 −20	0 −29	0 −46	0 −72	0 −115	0 −185	±10	±14.5
250	315	0 −35	−190 −222	−110 −142	−56 −88	−17 −40	−17 −49	0 −23	0 −32	0 −52	0 −81	0 −130	0 −210	±11.5	±16
315	400	0 −40	−210 −246	−125 −161	−62 −98	−18 −43	−18 −54	0 −25	0 −36	0 −57	0 −89	0 −140	0 −230	±12.5	±18
400	500	0 −45	−230 −270	−135 −175	−68 −108	−20 −47	−20 −60	0 −27	0 −40	0 −63	0 −97	0 −155	0 −250	±13.5	±20
500	630	0 −50	−260 −304	−145 −189	−76 −120	—	−22 −66	—	0 −44	0 −70	0 −110	0 −175	0 −280	—	±22
630	800	0 −75	−290 −340	−160 −210	−80 −130	—	−24 −74	—	0 −50	0 −80	0 −125	0 −200	0 −320	—	±25
800	1 000	0 −100	−320 −376	−170 −226	−86 −142	—	−26 −82	—	0 −56	0 −90	0 −140	0 −230	0 −360	—	±28
1 000	1 250	0 −125	−350 −416	−195 −261	−98 −164	—	−28 −94	—	0 −66	0 −105	0 −165	0 −260	0 −420	—	±33
1 250	1 600	0 −160	−390 −468	−220 −298	−110 −188	—	−30 −108	—	0 −78	0 −125	0 −195	0 −310	0 −500	—	±39
1 600	2 000	0 −200	−430 −522	−240 −332	−120 −212	—	−32 −124	—	0 −92	0 −150	0 −230	0 −370	0 −600	—	±46

치 수 허 용 차

단위 μm

j5	j6	j7	k5	k6	k7	m5	m6	n6	p6	r6	r7	경의 구분(mm) 초과	이하
+3 −2	+6 −2	+8 −4	+6 +1	+9 +1	+13 +1	+9 +4	+12 +4	+16 +8	+20 +12	+23 +15	+27 +15	3	6
+4 −2	+7 −2	+10 −5	+7 +1	+10 +1	+16 +1	+12 +6	+15 +6	+19 +10	+24 +15	+28 +19	+34 +19	6	10
+5 −3	+8 −3	+12 −6	+9 +1	+12 +1	+19 +1	+15 +7	+18 +7	+23 +12	+29 +18	+34 +23	+41 +23	10	18
+5 −4	+9 −4	+13 −8	+11 +2	+15 +2	+23 +2	+17 +8	+21 +8	+28 +15	+35 +22	+41 +28	+49 +28	18	30
+6 −5	+11 −5	+15 −10	+13 +2	+18 +2	+27 +2	+20 +9	+25 +9	+33 +17	+42 +26	+50 +34	+59 +34	30	50
+6 −7	+12 −7	+18 −12	+15 +2	+21 +2	+32 +2	+24 +11	+30 +11	+39 +20	+51 +32	+60 +41	+71 +41	50	65
										+62 +43	+73 +43	65	80
+6 −9	+13 −9	+20 −15	+18 +3	+25 +3	+38 +3	+28 +13	+35 +13	+45 +23	+59 +37	+73 +51	+86 +51	80	100
										+76 +54	+89 +54	100	120
+7 −11	+14 −11	+22 −18	+21 +3	+28 +3	+43 +3	+33 +15	+40 +15	+52 +27	+68 +43	+88 +63	+103 +63	120	140
										+90 +65	+105 +65	140	160
										+93 +68	+108 +68	160	180
+7 −13	+16 −13	+25 −21	+24 +4	+33 +4	+50 +4	+37 +17	+46 +17	+60 +31	+79 +50	+106 +77	+123 +77	180	200
										+109 +80	+126 +80	200	225
										+113 +84	+130 +84	225	250
+7 −16	±16	±26	+27 +4	+36 +4	+56 +4	+43 +20	+52 +20	+66 +34	+88 +56	+126 +94	+146 +94	250	280
										+130 +98	+150 +98	280	315
+7 −18	±18	+29 −28	+29 +4	+40 +4	+61 +4	+46 +21	+57 +21	+73 +37	+98 +62	+144 +108	+165 +108	315	355
										+150 +114	+171 +114	355	400
+7 −20	±20	+31 −32	+32 +5	+45 +5	+68 +5	+50 +23	+63 +23	+80 +40	+108 +68	+166 +126	+189 +126	400	450
										+172 +132	+195 +132	450	500
—	—	—	—	+44 0	+70 0	—	+70 +26	+88 +44	+122 +78	+194 +150	+220 +150	500	560
										+199 +155	+225 +155	560	630
—	—	—	—	+50 0	+80 0	—	+80 +30	+100 +50	+138 +88	+225 +175	+255 +175	630	710
										+235 +185	+265 +185	710	800
—	—	—	—	+56 0	+90 0	—	+90 +34	+112 +56	+156 +100	+266 +210	+300 +210	800	900
										+276 +220	+310 +220	900	1 000
—	—	—	—	+66 0	+105 0	—	+106 +40	+132 +66	+186 +120	+316 +250	+355 +250	1 000	1 120
										+326 +260	+365 +260	1 120	1 250
—	—	—	—	+78 0	+125 0	—	+126 +48	+156 +78	+218 +140	+378 +300	+425 +300	1 250	1 400
										+408 +330	+455 +330	1 400	1 600
—	—	—	—	+92 0	+150 0	—	+150 +58	+184 +92	+262 +170	+462 +370	+520 +370	1 600	1 800
										+492 +400	+550 +400	1 800	2 000

부 착 10 하우징구멍의

경의 구분(mm) 초과	이하	베어링의 평면 내평균외경의 치수차(0급) Δ_{Dmp}	E6	F6	F7	G6	G7	H6	H7	H8	J6	J7	JS6	JS7
10	18	0 − 8	+ 43 + 32	+ 27 + 16	+ 34 + 16	+ 17 + 6	+ 24 + 6	+ 11 0	+ 18 0	+ 27 0	+ 6 − 5	+ 10 − 8	± 5.5	± 9
18	30	0 − 9	+ 53 + 40	+ 33 + 20	+ 41 + 20	+ 20 + 7	+ 28 + 7	+ 13 0	+ 21 0	+ 33 0	+ 8 − 5	+ 12 − 9	± 6.5	± 10.5
30	50	0 − 11	+ 66 + 50	+ 41 + 25	+ 50 + 25	+ 25 + 9	+ 34 + 9	+ 16 0	+ 25 0	+ 39 0	+ 10 − 6	+ 14 − 11	± 8	± 12.5
50	80	0 − 13	+ 79 + 60	+ 49 + 30	+ 60 + 30	+ 29 + 10	+ 40 + 10	+ 19 0	+ 30 0	+ 46 0	+ 13 − 6	+ 18 − 12	± 9.5	± 15
80	120	0 − 15	+ 94 + 72	+ 58 + 36	+ 71 + 36	+ 34 + 12	+ 47 + 12	+ 22 0	+ 35 0	+ 54 0	+ 16 − 6	+ 22 − 13	± 11	± 17.5
120 150	150 180	0 − 18 0 − 25	+ 110 + 85	+ 68 + 43	+ 83 + 43	+ 39 + 14	+ 54 + 14	+ 25 0	+ 40 0	+ 63 0	+ 18 − 7	+ 26 − 14	± 12.5	± 20
180	250	0 − 30	+ 129 + 100	+ 79 + 50	+ 96 + 50	+ 44 + 15	+ 61 + 15	+ 29 0	+ 46 0	+ 72 0	+ 22 − 7	+ 30 − 16	± 14.5	± 23
250	315	0 − 35	+ 142 + 110	+ 88 + 56	+ 108 + 56	+ 49 + 17	+ 69 + 17	+ 32 0	+ 52 0	+ 81 0	+ 25 − 7	+ 36 − 16	± 16	± 26
315	400	0 − 40	+ 161 + 125	+ 98 + 62	+ 119 + 62	+ 54 + 18	+ 75 + 18	+ 36 0	+ 57 0	+ 89 0	+ 29 − 7	+ 39 − 18	± 18	± 28.5
400	500	0 − 45	+ 175 + 135	+ 108 + 68	+ 131 + 68	+ 60 + 20	+ 83 + 20	+ 40 0	+ 63 0	+ 97 0	+ 33 − 7	+ 43 − 20	± 20	± 31.5
500	630	0 − 50	+ 189 + 145	+ 120 + 76	+ 146 + 76	+ 66 + 22	+ 92 + 22	+ 44 0	+ 70 0	+ 110 0	—	—	± 22	± 35
630	800	0 − 75	+ 210 + 160	+ 130 + 80	+ 160 + 80	+ 74 + 24	+ 104 + 24	+ 50 0	+ 80 0	+ 125 0	—	—	± 25	± 40
800	1 000	0 − 100	+ 226 + 170	+ 142 + 86	+ 176 + 86	+ 82 + 26	+ 116 + 26	+ 56 0	+ 90 0	+ 140 0	—	—	± 28	± 45
1 000	1 250	0 − 125	+ 261 + 195	+ 164 + 98	+ 203 + 98	+ 94 + 28	+ 133 + 28	+ 66 0	+ 105 0	+ 165 0	—	—	± 33	± 52.5
1 250	1 600	0 − 160	+ 298 + 220	+ 188 + 110	+ 235 + 110	+ 108 + 30	+ 155 + 30	+ 78 0	+ 125 0	+ 195 0	—	—	± 39	± 62.5
1 600	2 000	0 − 200	+ 332 + 240	+ 212 + 120	+ 270 + 120	+ 124 + 32	+ 182 + 32	+ 92 0	+ 150 0	+ 230 0	—	—	± 46	± 75
2 000	2 500	0 − 250	+ 370 + 260	+ 240 + 130	+ 305 + 130	+ 144 + 34	+ 209 + 34	+ 110 0	+ 175 0	+ 280 0	—	—	± 55	± 87.5

치수허용차

단위 μm

K5	K6	K7	M5	M6	M7	N5	N6	N7	P6	P7	경의 구분(mm) 초과	경의 구분(mm) 이하
+2 −6	+2 −9	+6 −12	−4 −12	−4 −15	0 −18	−9 −17	−9 −20	−5 −23	−15 −26	−11 −29	10	18
+1 −8	+2 −11	+6 −15	−5 −14	−4 −17	0 −21	−12 −21	−11 −24	−7 −28	−18 −31	−14 −35	18	30
+2 −9	+3 −13	+7 −18	−5 −16	−4 −20	0 −25	−13 −24	−12 −28	−8 −33	−21 −37	−17 −42	30	50
+3 −10	+4 −15	+9 −21	−6 −19	−5 −24	0 −30	−15 −28	−14 −33	−9 −39	−26 −45	−21 −51	50	80
+2 −13	+4 −18	+10 −25	−8 −23	−6 −28	0 −35	−18 −33	−16 −38	−10 −45	−30 −52	−24 −59	80	120
+3 −15	+4 −21	+12 −28	−9 −27	−8 −33	0 −40	−21 −39	−20 −45	−12 −52	−36 −61	−28 −68	120	180
+2 −18	+5 −24	+13 −33	−11 −31	−8 −37	0 −46	−25 −45	−22 −51	−14 −60	−41 −70	−33 −79	180	250
+3 −20	+5 −27	+16 −36	−13 −36	−9 −41	0 −52	−27 −50	−25 −57	−14 −66	−47 −79	−36 −88	250	315
+3 −22	+7 −29	+17 −40	−14 −39	−10 −46	0 −57	−30 −55	−26 −62	−16 −73	−51 −87	−41 −98	315	400
+2 −25	+8 −32	+18 −45	−16 −43	−10 −50	0 −63	−33 −60	−27 −67	−17 −80	−55 −95	−45 −108	400	500
—	0 −44	0 −70	—	−26 −70	−26 −96	—	−44 −88	−44 −114	−78 −122	−78 −148	500	630
—	0 −50	0 −80	—	−30 −80	−30 −110	—	−50 −100	−50 −130	−88 −138	−88 −168	630	800
—	0 −56	0 −90	—	−34 −90	−34 −124	—	−56 −112	−56 −146	−100 −156	−100 −190	800	1 000
—	0 −66	0 −105	—	−40 −106	−40 −145	—	−66 −132	−66 −171	−120 −186	−120 −225	1 000	1 250
—	0 −78	0 −125	—	−48 −126	−48 −173	—	−78 −156	−78 −203	−140 −218	−140 −265	1 250	1 600
—	0 −92	0 −150	—	−58 −150	−58 −208	—	−92 −184	−92 −242	−170 −262	−170 −320	1 600	2 000
—	0 −110	0 −175	—	−68 −178	−68 −243	—	−110 −220	−110 −285	−195 −305	−195 −370	2 000	2 500

부 착 11 기본공차 IT

기준치수의 구 분 (mm)		공차										
		IT1	IT2	IT3	IT4	IT5	IT6	IT7	IT8	IT9	IT10	IT11
초과	이하	기본공차의 수치 (μm)										
—	3	0.8	1.2	2	3	4	6	10	14	25	40	60
3	6	1	1.5	2.5	4	5	8	12	18	30	48	75
6	10	1	1.5	2.5	4	6	9	15	22	36	58	90
10	18	1.2	2	3	5	8	11	18	27	43	70	110
18	30	1.5	2.5	4	6	9	13	21	33	52	84	130
30	50	1.5	2.5	4	7	11	16	25	39	62	100	160
50	80	2	3	5	8	13	19	30	46	74	120	190
80	120	2.5	4	6	10	15	22	35	54	87	140	220
120	180	3.5	5	8	12	18	25	40	63	100	160	250
180	250	4.5	7	10	14	20	29	46	72	115	185	290
250	315	6	8	12	16	23	32	52	81	130	210	320
315	400	7	9	13	18	25	36	57	89	140	230	360
400	500	8	10	15	20	27	40	63	97	155	250	400
500	630	9	11	16	22	32	44	70	110	175	280	440
630	800	10	13	18	25	36	50	80	125	200	320	500
800	1 000	11	15	21	28	40	56	90	140	230	360	560
1 000	1 250	13	18	24	33	47	66	105	165	260	420	660
1 250	1 600	15	21	29	39	55	78	125	195	310	500	780
1 600	2 000	18	25	35	46	65	92	150	230	370	600	920
2 000	2 500	22	30	41	55	78	110	175	280	440	700	1 100
2 500	3 150	26	36	50	68	96	135	210	330	540	860	1 350

비 고 1. 공차등급 IT14~IT18은, 기준치수 1mm이하에는 적용하지 않는다.
2. 500mm를 초과하는 기준치수에 대한 공차등급 IT1~IT5의 공차의 값은 실험적으로 사용하기 위한 暫定的인 것이다.

의 수치

등급							기준공차의 구분 (mm)	
IT12	IT13	IT14	IT15	IT16	IT17	IT18		
기본공차의 수치 (mm)							초과	이하
0.10	0.14	0.25	0.40	0.60	1.00	1.40	—	3
0.12	0.18	0.30	0.48	0.75	1.20	1.80	3	6
0.15	0.22	0.36	0.58	0.90	1.50	2.20	6	10
0.18	0.27	0.43	0.70	1.10	1.80	2.70	10	18
0.21	0.33	0.52	0.84	1.30	2.10	3.30	18	30
0.25	0.39	0.62	1.00	1.60	2.50	3.90	30	50
0.30	0.46	0.74	1.20	1.90	3.00	4.60	50	80
0.35	0.54	0.87	1.40	2.20	3.50	5.40	80	120
0.40	0.63	1.00	1.60	2.50	4.00	6.30	120	180
0.46	0.72	1.15	1.85	2.90	4.60	7.20	180	250
0.52	0.81	1.30	2.10	3.20	5.20	8.10	250	315
0.57	0.89	1.40	2.30	3.60	5.70	8.90	315	400
0.63	0.97	1.55	2.50	4.00	6.30	9.70	400	500
0.70	1.10	1.75	2.80	4.40	7.00	11.00	500	630
0.80	1.25	2.00	3.20	5.00	8.00	12.50	630	800
0.90	1.40	2.30	3.60	5.60	9.00	14.00	800	1 000
1.05	1.65	2.60	4.20	6.60	10.50	16.50	1 000	1 250
1.25	1.95	3.10	5.00	7.80	12.50	19.50	1 250	1 600
1.50	2.30	3.70	6.00	9.20	15.00	23.00	1 600	2 000
1.75	2.80	4.40	7.00	11.00	17.50	28.00	2 000	2 500
2.10	3.30	5.40	8.60	13.50	21.00	33.00	2 500	3 150

부표 12 회전속도 n과 속도계수 f_n

볼 베 어 링 $f_n = (0.03n)^{-1/3}$

로울러베어링 $f_n = (0.03n)^{-3/10}$

회전속도 n (min^{-1})	속 도 계 수 f_n	
	볼베어링	로울러베어링
10	1.49	1.44
11	1.45	1.39
12	1.41	1.36
13	1.37	1.33
14	1.34	1.30
15	1.30	1.27
16	1.28	1.25
17	1.25	1.22
18	1.23	1.20
19	1.21	1.18
20	1.19	1.17
21	1.17	1.15
22	1.15	1.13
23	1.13	1.12
24	1.12	1.10
25	1.10	1.09
26	1.09	1.08
27	1.07	1.07
28	1.06	1.05
29	1.05	1.04
30	1.04	1.03
31	1.02	1.02
32	1.01	1.01
33.3	**1.00**	**1.00**
34	0.993	0.994
36	0.975	0.977
38	0.957	0.961
40	0.941	0.947
42	0.926	0.933
44	0.912	0.920
46	0.898	0.908
48	0.886	0.896
50	0.874	0.885
55	0.846	0.861
60	0.822	0.838
65	0.800	0.818
70	0.781	0.800
75	0.763	0.784
80	0.747	0.769
85	0.732	0.755
90	0.718	0.742
95	0.705	0.730
100	0.693	0.719
110	0.672	0.699
120	0.652	0.681
130	0.635	0.665
140	0.620	0.650
150	0.606	0.637
160	0.593	0.625
170	0.581	0.613

회전속도 n (min^{-1})	속 도 계 수 f_n	
	볼베어링	로울러베어링
180	0.570	0.603
190	0.560	0.593
200	0.550	0.584
220	0.533	0.568
240	0.518	0.553
260	0.504	0.540
280	0.492	0.528
300	0.481	0.517
320	0.471	0.507
340	0.461	0.498
360	0.452	0.490
380	0.444	0.482
400	0.437	0.475
420	0.430	0.468
440	0.423	0.461
460	0.417	0.455
480	0.411	0.449
500	0.405	0.444
550	0.393	0.431
600	0.382	0.420
650	0.372	0.410
700	0.362	0.401
750	0.354	0.393
800	0.347	0.385
850	0.340	0.378
900	0.333	0.372
950	0.327	0.366
1 000	0.322	0.360
1 050	0.317	0.355
1 100	0.312	0.350
1 150	0.307	0.346
1 200	0.303	0.341
1 250	0.299	0.337
1 300	0.295	0.333
1 400	0.288	0.326
1 500	0.281	0.319
1 600	0.275	0.313
1 700	0.270	0.307
1 800	0.265	0.302
1 900	0.260	0.297
2 000	0.255	0.293
2 100	0.251	0.289
2 200	0.247	0.285
2 300	0.244	0.281
2 400	0.240	0.277
2 500	0.237	0.274
2 600	0.234	0.271
2 700	0.231	0.268
2 800	0.228	0.265
2 900	0.226	0.262

회전속도 n (min^{-1})	속 도 계 수 f_n	
	볼베어링	로울러베어링
3 000	0.223	0.259
3 200	0.218	0.254
3 400	0.214	0.250
3 600	0.210	0.245
3 800	0.206	0.242
4 000	0.203	0.238
4 200	0.199	0.234
4 400	0.196	0.231
4 600	0.194	0.228
4 800	0.191	0.225
5 000	0.188	0.222
5 200	0.186	0.220
5 400	0.183	0.217
5 600	0.181	0.215
5 800	0.179	0.213
6 000	0.177	0.211
6 200	0.175	0.209
6 400	0.173	0.207
6 600	0.172	0.205
6 800	0.170	0.203
7 000	0.168	0.201
7 200	0.167	0.199
7 400	0.165	0.198
7 600	0.164	0.196
7 800	0.162	0.195
8 000	0.161	0.193
8 500	0.158	0.190
9 000	0.155	0.186
9 500	0.152	0.183
10 000	0.149	0.181
11 000	0.145	0.176
12 000	0.141	0.171
13 000	0.137	0.167
14 000	0.134	0.163
15 000	0.130	0.160
16 000	0.128	0.157
17 000	0.125	0.154
18 000	0.123	0.151
19 000	0.121	0.149
20 000	0.119	0.147
22 000	0.115	0.143
24 000	0.112	0.139
26 000	0.109	0.136
28 000	0.106	0.133
30 000	0.104	0.130
32 000	0.101	0.127
34 000	0.099	0.125
36 000	0.097	0.123
38 000	0.096	0.121
40 000	0.094	0.119

부표 13 하중비 C/P · 피로수명계수 f_n 와 피로수명 $L \cdot L_h$

볼베어링 $L=(C/P)^3$ $L_h=500f_n^3$
로울러베어링 $L=(C/P)^{10/3}$ $L_h=500f_n^{10/3}$

하중비 C/P 또는 피로수명계수 f_h	볼베어링의 피로수명		로울러베어링의 피로수명	
	L (10^6rev)	L_h (h)	L (10^6rev)	L_h (h)
0.70	0.34	172	0.30	152
0.75	0.42	211	0.38	192
0.80	0.51	256	0.48	238
0.85	0.61	307	0.58	291
0.90	0.73	365	0.70	352
0.95	0.86	429	0.84	421
1.00	**1.00**	**500**	**1.00**	**500**
1.05	1.16	579	1.18	588
1.10	1.33	665	1.37	687
1.15	1.52	760	1.59	797
1.20	1.73	864	1.84	918
1.25	1.95	977	2.10	1 050
1.30	2.20	1 100	2.40	1 200
1.35	2.46	1 230	2.72	1 360
1.40	2.74	1 370	3.07	1 530
1.45	3.05	1 520	3.45	1 730
1.50	3.38	1 690	3.86	1 930
1.55	3.72	1 860	4.31	2 150
1.60	4.10	2 050	4.79	2 400
1.65	4.49	2 250	5.31	2 650
1.70	4.91	2 460	5.86	2 930
1.75	5.36	2 680	6.46	3 230
1.80	5.83	2 920	7.09	3 550
1.85	6.33	3 170	7.77	3 890
1.90	6.86	3 430	8.50	4 250
1.95	7.41	3 710	9.26	4 630
2.00	8.00	4 000	10.1	5 040
2.05	8.62	4 310	10.9	5 470
2.10	9.26	4 630	11.9	5 930
2.15	9.94	4 970	12.8	6 410
2.20	10.6	5 320	13.8	6 920
2.25	11.4	5 700	14.9	7 460
2.30	12.2	6 080	16.1	8 030
2.35	13.0	6 490	17.3	8 630
2.40	13.8	6 910	18.5	9 250
2.45	14.7	7 350	19.8	9 910
2.50	15.6	7 810	21.2	10 600
2.55	16.6	8 290	22.7	11 300
2.60	17.6	8 790	24.2	12 100
2.65	18.6	9 300	25.8	12 900
2.70	19.7	9 840	27.4	13 700
2.75	20.8	10 400	29.1	14 600
2.80	22.0	11 000	30.9	15 500
2.85	23.1	11 600	32.8	16 400
2.90	24.4	12 200	34.8	17 400
2.95	25.7	12 800	36.8	18 400
3.00	27.0	13 500	38.9	19 500
3.05	28.4	14 200	41.1	20 600
3.10	29.8	14 900	43.4	21 700
3.15	31.3	15 600	45.8	22 900
3.20	32.8	16 400	48.3	24 100
3.25	34.3	17 200	50.8	25 400
3.30	35.9	18 000	53.5	26 800
3.35	37.6	18 800	56.3	28 100
3.40	39.3	19 700	59.1	29 600

하중비 C/P 또는 피로수명계수 f_h	볼베어링의 피로수명		로울러베어링의 피로수명	
	L (10^6rev)	L_h (h)	L (10^6rev)	L_h (h)
3.45	41.1	20 500	62.0	31 000
3.50	42.9	21 400	65.1	32 500
3.55	44.7	22 400	68.2	34 100
3.60	46.7	23 300	71.5	35 800
3.65	48.6	24 300	74.9	37 400
3.70	50.7	25 300	78.3	39 200
3.75	52.7	26 400	81.9	41 000
3.80	54.9	27 400	85.6	42 800
3.85	57.1	28 500	89.4	44 700
3.90	59.3	29 700	93.4	46 700
3.95	61.6	30 800	97.4	48 700
4.00	64.0	32 000	102	50 800
4.05	66.4	33 200	106	52 900
4.10	68.9	34 500	110	55 200
4.15	71.5	35 700	115	57 400
4.20	74.1	37 000	120	59 800
4.25	76.8	38 400	124	62 200
4.30	79.5	39 800	129	64 600
4.35	82.3	41 200	134	67 200
4.40	85.2	42 600	140	69 800
4.45	88.1	44 100	145	72 500
4.50	91.1	45 600	150	75 200
4.55	94.2	47 100	156	78 000
4.60	97.3	48 700	162	80 900
4.65	101	50 300	168	83 900
4.70	104	51 900	174	87 000
4.75	107	53 600	180	90 100
4.80	111	55 300	187	93 300
4.85	114	57 000	193	96 600
4.90	118	58 800	200	99 900
4.95	121	60 600	207	103 000
5.00	125	62 500	214	107 000
5.10	133	66 300	228	114 000
5.20	141	70 300	244	122 000
5.30	149	74 400	260	130 000
5.40	157	78 700	276	138 000
5.50	166	83 200	294	147 000
5.60	176	87 800	312	156 000
5.70	185	92 600	331	165 000
5.80	195	97 600	351	175 000
5.90	205	103 000	371	186 000
6.00	216	108 000	392	196 000
6.50	275	137 000	513	256 000
7.00	343	172 000	656	328 000
7.50	422	211 000	826	413 000
8.00	512	256 000	1 020	512 000
8.50	614	307 000	1 250	627 000
9.00	729	365 000	1 520	758 000
9.50	857	429 000	1 820	908 000
10.0	1 000	—	2 150	—
11.0	1 330	—	2 960	—
12.0	1 730	—	3 960	—
13.0	2 200	—	5 170	—
14.0	2 740	—	6 610	—
15.0	3 380	—	8 320	—

부 표 14 인치계열 테이퍼로울러베어링의 색인

호칭번호 CONE, CUP	호칭치수(mm) *d*:CONE 내경 *D*:CUP 외경		베어링記載페이지
332	*D*	80.000	B136,B140,B142
336	*d*	41.275	B142
342	*d*	41.275	B142
342 S	*d*	42.875	B142
344	*d*	40.000	B140
344 A	*d*	40.000	B140
346	*d*	31.750	B136
354 A	*D*	85.000	B144
359 S	*d*	46.038	B144
362 A	*D*	88.900	B144,B146
366	*d*	50.000	B146
368	*d*	50.800	B146
368 A	*d*	50.800	B146
369 A	*d*	47.625	B144
372	*D*	100.000	B146
374	*D*	93.264	B144
376	*d*	45.000	B144
377	*d*	52.388	B146
382	*D*	98.425	B148
382 A	*D*	96.838	B148
382 S	*D*	96.838	B148
385	*d*	55.000	B148
387	*d*	57.150	B148
387 A	*d*	57.150	B148
388 A	*d*	57.531	B148
390 A	*d*	63.500	B150
394 A	*D*	110.000	B150,B152
395	*d*	63.500	B150
395 A	*d*	66.675	B152
395 S	*d*	66.675	B152
397	*d*	60.000	B150
399 A	*d*	68.262	B152
414	*D*	88.501	B140
418	*d*	38.100	B140
432	*D*	95.250	B142
432 A	*D*	95.250	B144
436	*d*	46.038	B144
438	*d*	44.450	B142
453 A	*D*	107.950	B144
453 X	*D*	104.775	B148
460	*d*	44.450	B144
462	*d*	57.150	B148
469	*d*	57.150	B148
472	*D*	120.000	B152,B154
472 A	*D*	120.000	B152
478	*d*	65.000	B152
480	*d*	68.262	B152
484	*d*	70.000	B154
492 A	*D*	133.350	B156,B158
493	*D*	136.525	B154,B156,B158
495	*d*	82.550	B156
495 A	*d*	76.200	B154
495 AX	*d*	76.200	B154
496	*d*	80.962	B156

호칭번호 CONE, CUP	호칭치수(mm) *d*:CONE 내경 *D*:CUP 외경		베어링記載페이지
497	*d*	85.725	B158
498	*d*	84.138	B158
522	*D*	101.600	B144,B146
528	*d*	47.625	B144
529	*d*	50.800	B146
529 X	*d*	50.800	B146
532 X	*D*	107.950	B148
539	*d*	53.975	B148
552 A	*D*	123.825	B148,B150,B152
553 X	*D*	122.238	B150,B152
555 S	*d*	57.150	B148
557 S	*d*	53.975	B148
558	*d*	60.325	B150
559	*d*	63.500	B150
560	*d*	66.675	B152
560 S	*d*	68.262	B152
563	*D*	127.000	B150,B152,B154
563 X	*D*	127.000	B152
565	*d*	63.500	B150
566	*d*	69.850	B152
567	*d*	73.025	B154
567 A	*d*	71.438	B154
567 S	*d*	71.438	B154
568	*d*	73.817	B154
569	*d*	64.963	B150
570	*d*	68.262	B152
572	*D*	139.992	B154,B156
572 X	*D*	139.700	B156
575	*d*	76.200	B154
580	*d*	82.550	B156
581	*d*	80.962	B156
582	*d*	82.550	B156
590 A	*d*	76.200	B154
592	*D*	152.400	B160
592 A	*D*	152.400	B154,B158,B160
593	*d*	88.900	B158
594	*d*	95.250	B160
596	*d*	85.725	B158
597	*d*	93.662	B160
598	*d*	92.075	B160
598 A	*d*	92.075	B160
614 X	*D*	115.000	B148
622 X	*d*	55.000	B148
632	*D*	136.525	B150,B154
633	*D*	130.175	B150,B152,B154
637	*d*	60.325	B150
639	*d*	63.500	B150
643	*d*	69.850	B152
644	*d*	71.438	B154
645	*d*	71.438	B154
652	*D*	152.400	B154,B156
653	*D*	146.050	B152,B154,B156,B158
653 X	*D*	150.000	B154
655	*d*	69.850	B152

호칭번호 CONE, CCUP	호칭치수(mm) *d*:CONE 내경 *D*:CUP 외경		베어링記載페이지
657	*d*	73.025	B154
658	*d*	74.612	B154
659	*d*	76.200	B154
661	*d*	79.375	B156
663	*d*	82.550	B156
664	*d*	84.138	B158
665	*d*	85.725	B158
665 A	*d*	85.725	B158
672	*D*	168.275	B158,B160,B162
677	*d*	85.725	B158
681	*d*	92.075	B160
683	*d*	95.250	B160
685	*d*	98.425	B160
687	*d*	101.600	B162
742	*D*	150.089	B152,B156,B158
743	*D*	150.000	B156
745 A	*d*	69.850	B152
749	*d*	85.026	B158
749 A	*d*	82.550	B156
749 S	*d*	85.026	B158
750	*d*	79.375	B156
752	*D*	161.925	B156,B158
753	*D*	168.275	B156,B158
757	*d*	82.550	B156
758	*d*	85.725	B158
759	*d*	88.900	B158
760	*d*	90.488	B158
766	*d*	88.900	B158
772	*D*	180.975	B160,B162
776	*d*	95.250	B160
779	*d*	98.425	B160
780	*d*	101.600	B162
782	*d*	104.775	B162
787	*d*	104.775	B162
792	*D*	206.375	B164
795	*d*	120.650	B164
797	*d*	130.000	B164
799	*d*	128.588	B164
799 A	*d*	130.175	B164
832	*D*	168.275	B156,B158
837	*d*	76.200	B156
842	*d*	82.550	B156
843	*d*	76.200	B156
850	*d*	88.900	B158
854	*D*	190.500	B158,B160,B162
855	*d*	88.900	B158
857	*d*	92.075	B160
861	*d*	101.600	B162
864	*d*	95.250	B160
866	*d*	98.425	B160
932	*D*	212.725	B162
938	*d*	114.300	B162
1220	*D*	57.150	B132
1280	*d*	22.225	B132

호칭번호 CONE, CCUP	호칭치수(mm) *d*:CONE 내경 *D*:CUP 외경		베어링記載페이지
1328	*D*	52.388	B132
1329	*D*	53.975	B132
1380	*d*	22.225	B132
1620	*D*	66.675	B138
1680	*d*	33.338	B138
1729	*D*	56.896	B132,B134
1755	*d*	22.225	B132
1779	*d*	23.812	B134
1922	*D*	57.150	B134
1988	*d*	28.575	B134
1997 X	*d*	26.988	B134
A2047	*d*	12.000	B132
A2126	*D*	31.991	B132
2523	*D*	69.850	B136,B138
2558	*d*	30.162	B136
2559	*d*	30.162	B136
2580	*d*	31.750	B136
2582	*d*	31.750	B136
2585	*d*	33.338	B138
2631	*D*	66.421	B136
2690	*d*	29.367	B136
2720	*D*	76.200	B140
2729	*D*	76.200	B140
2735 X	*D*	73.025	B140
2788	*d*	38.100	B140
2789	*d*	39.688	B140
2820	*D*	73.025	B138
2877	*d*	34.925	B138
2924	*D*	85.000	B144
2984	*d*	46.038	B144
3120	*D*	72.626	B136,B138
3188	*d*	31.750	B136
3197	*d*	33.338	B138
3320	*D*	80.167	B140
3386	*d*	39.688	B140
3420	*D*	79.375	B138,B140
3478	*d*	34.925	B138
3479	*d*	36.512	B140
3490	*d*	38.100	B140
3525	*D*	87.312	B142
3576	*d*	41.275	B142
3578	*d*	44.450	B142
3720	*D*	93.264	B142
3730	*D*	93.264	B146
3775	*d*	50.800	B146
3780	*d*	50.800	B146
3782	*d*	44.450	B142
3820	*D*	85.725	B142
3877	*d*	41.275	B142
3920	*D*	112.712	B150,B152
3926	*D*	112.712	B148,B150
3981	*d*	58.738	B148
3982	*d*	63.500	B150
3984	*d*	66.675	B152

호칭번호 CONE, CCUP	호칭치수(mm) d:CONE 내경 D:CUP 외경		베어링記載페이지
3994	*d*	66.675	B152
A4050	*d*	12.700	B132
A4059	*d*	15.000	B132
A4138	*D*	34.988	B132
4335	*D*	90.488	B142
4388	*d*	41.275	B142
4535	*D*	104.775	B148
4595	*d*	53.975	B148
A5069	*d*	17.455	B132
A5144	*D*	36.525	B132
5335	*D*	103.188	B144
5356	*d*	44.450	B144
5535	*D*	122.238	B148,B150
5566	*d*	55.562	B148
5582	*d*	60.325	B150
5584	*d*	63.500	B150
5735	*D*	135.733	B154,B156
5760	*d*	76.200	B154
5795	*d*	77.788	B156
A6062	*d*	15.875	B132
A6067	*d*	16.993	B132
A6075	*d*	19.050	B132
A6157	*D*	39.992	B132
6220	*D*	127.000	B146,B148
6279	*d*	50.800	B146
6280	*d*	53.975	B148
6320	*D*	135.755	B150,B152
6376	*d*	60.325	B150
6379	*d*	65.088	B152
6420	*D*	149.225	B148,B152,B154
6454	*d*	69.850	B152
6455	*d*	57.150	B148
6460	*d*	73.025	B154
6461	*d*	76.200	B154
6535	*D*	161.925	B154,B156,B158
6536	*D*	161.925	B154
6559	*d*	82.550	B156
6575	*d*	76.200	B154
6576	*d*	76.200	B154
6580	*d*	88.900	B158
9121	*D*	152.400	B150,B152
9180	*d*	61.912	B150
9185	*d*	68.262	B152
9220	*D*	161.925	B154
9285	*d*	76.200	B154
9320	*D*	177.800	B156
9321	*D*	171.450	B156,B158
9378	*d*	76.200	B156
9380	*d*	76.200	B156
9385	*d*	84.138	B158
02420	*D*	68.262	B134,B136
02473	*d*	25.400	B134
02474	*d*	28.575	B134
02475	*d*	31.750	B136

호칭번호 CONE, CCUP	호칭치수(mm) d:CONE 내경 D:CUP 외경		베어링記載페이지
02820	*D*	73.025	B134,B138
02872	*d*	28.575	B134
02878	*d*	34.925	B138
03062	*d*	15.875	B132
03162	*D*	41.275	B132
05062	*d*	15.875	B132
05068	*d*	17.462	B132
05075	*d*	19.050	B132
05079	*d*	19.990	B132
05175	*D*	44.450	B132
05185	*D*	47.000	B132
07079	*d*	20.000	B132
07087	*d*	22.225	B132
07097	*d*	25.000	B134
07098	*d*	24.981	B134
07100	*d*	25.400	B134
07100SA	*d*	25.400	B134
07196	*D*	50.005	B132,B134
07204	*D*	51.994	B132,B134
07205	*D*	52.001	B134
08118	*d*	30.162	B136
08125	*d*	31.750	B136
08231	*D*	58.738	B136
09062	*d*	15.875	B132
09067	*d*	19.050	B132
09074	*d*	19.050	B132
09078	*d*	19.050	B132
09081	*d*	20.625	B132
09194	*D*	49.225	B132
09195	*D*	49.225	B132
09196	*D*	49.225	B132
11162	*d*	41.275	B142
11300	*D*	76.200	B142
11520	*D*	42.862	B132
11590	*d*	15.875	B132
LM11710	*D*	39.878	B132
LM11749	*d*	17.462	B132
LM11910	*D*	45.237	B132
LM11949	*d*	19.050	B132
12168	*d*	42.862	B142
12303	*D*	76.992	B142
12520	*D*	49.225	B132
12580	*d*	20.638	B132
M12610	*D*	50.005	B132
M12648	*d*	22.225	B132
M12649	*d*	21.430	B132
LM12710	*D*	45.237	B132
LM12711	*D*	45.975	B132
LM12749	*d*	22.000	B132
13175	*d*	44.450	B142
13181	*d*	46.038	B144
13318	*D*	80.962	B142,B144
13620	*D*	69.012	B140
13621	*D*	69.012	B140

호칭번호 CONE, CCUP	호칭치수(mm) d:CONE 내경 D:CUP 외경		베어링記載페이지
13685	*d*	38.100	B140
13687	*d*	38.100	B140
13830	*D*	63.500	B140
13889	*d*	38.100	B140
14123 A	*d*	31.750	B136
14125 A	*d*	31.750	B136
14130	*d*	33.338	B138
14131	*d*	33.338	B138
14137 A	*d*	34.925	B138
14138 A	*d*	34.925	B138
14139	*d*	34.976	B138
14274	*D*	69.012	B136,B138
14276	*D*	69.012	B136,B138
14283	*D*	72.085	B138
15100	*d*	25.400	B134
15101	*d*	25.400	B134
15106	*d*	26.988	B134
15112	*d*	28.575	B134
15113	*d*	28.575	B134
15116	*d*	30.112	B136
15117	*d*	30.000	B136
15118	*d*	30.213	B136
15119	*d*	30.213	B136
15120	*d*	30.213	B136
15123	*d*	31.750	B136
15125	*d*	31.750	B136
15126	*d*	31.750	B136
15245	*D*	62.000	B134,B136
15250	*D*	63.500	B136
15250 X	*D*	63.500	B134
15520	*D*	57.150	B134
15523	*D*	60.325	B134
15578	*d*	25.400	B134
15580	*d*	26.988	B134
16150	*d*	38.100	B140
16284	*D*	72.238	B140
16929	*D*	74.988	B142
16986	*d*	43.000	B142
17098	*d*	24.981	B134
17118	*d*	30.000	B136
17244	*D*	62.000	B134,B136
17520	*D*	42.862	B132
17580	*d*	15.875	B132
17831	*D*	79.985	B144
17887	*d*	45.230	B144
18200	*d*	50.800	B146
18337	*D*	85.725	B146
18520	*D*	73.025	B140
18590	*d*	41.275	B140
18620	*D*	79.375	B144
18690	*d*	46.038	B144
18720	*D*	85.000	B146
18790	*d*	50.800	B146
19138	*d*	34.976	B138

호칭번호 CONE, CCUP	호칭치수(mm) d:CONE 내경 D:CUP 외경		베어링記載페이지
19150	*d*	38.100	B140
19268	*D*	68.262	B138,B140
21075	*d*	19.050	B132
21212	*D*	53.975	B132
L21511	*D*	34.988	B132
L21549	*d*	15.875	B132
22168	*d*	42.862	B142
22325	*D*	82.550	B142
23100	*d*	25.400	B134
23256	*D*	65.088	B134
23621	*D*	73.025	B138
23691	*d*	35.000	B138
24720	*D*	76.200	B142
24721	*D*	76.200	B142
24780	*d*	41.275	B142
25520	*D*	82.931	B142,B144
25521	*D*	83.058	B142
25523	*D*	82.931	B142,B144
25577	*d*	42.875	B142
25578	*d*	42.862	B142
25580	*d*	44.450	B142
25584	*d*	44.983	B144
25590	*d*	45.618	B144
25820	*D*	73.025	B138
25821	*D*	73.025	B138,B140
25877	*d*	34.925	B138
25878	*d*	34.925	B138
25880	*d*	36.487	B140
26118	*d*	30.000	B136
26131	*d*	33.338	B138
26283	*D*	72.000	B136,B138
26820	*D*	80.167	B142
26822	*D*	79.375	B142
26823	*D*	76.200	B142
26882	*d*	41.275	B142
26884	*d*	42.875	B142
27620	*D*	125.412	B156
27687	*d*	82.550	B156
27689	*d*	83.345	B156
27690	*d*	83.345	B156
27820	*D*	80.035	B140
27880	*d*	38.100	B140
28138	*d*	34.976	B138
28315	*D*	80.000	B138
28521	*D*	92.075	B146
28580	*d*	50.800	B146
28584	*d*	52.388	B146
28622	*D*	97.630	B148
28680	*d*	55.562	B148
28920	*D*	101.600	B150
28921	*D*	100.000	B150
28985	*d*	60.325	B150
29520	*D*	107.950	B150
29586	*d*	63.500	B150

호칭번호 CONE, CCUP	호칭치수(mm) *d*:CONE 내경 *D*:CUP 외경		베어링記載페이지
29620	*D*	112.712	B152,B154
29630	*D*	120.650	B152
29675	*d*	69.850	B152
29685	*d*	73.025	B154
LM29710	*D*	65.088	B140
LM29711	*D*	65.088	B140
LM29748	*d*	38.100	B140
LM29749	*d*	38.100	B140
31520	*D*	76.200	B138
31594	*d*	34.925	B138
33262	*d*	66.675	B152
33275	*d*	69.850	B152
33281	*d*	71.438	B154
33287	*d*	73.025	B154
JHM33410	*D*	55.000	B134
JHM33449	*d*	24.000	B134
33462	*D*	117.475	B152,B154
33821	*D*	95.250	B146
33889	*d*	50.800	B146
34300	*d*	76.200	B154
34306	*d*	77.788	B156
34478	*D*	121.442	B154,B156
36620	*D*	193.675	B164
36690	*d*	146.050	B164
36920	*D*	227.012	B166
36990	*d*	177.800	B166
37425	*d*	107.950	B162
37625	*D*	158.750	B162
M38510	*D*	66.675	B138
M38511	*D*	65.987	B138
M38547	*d*	35.000	B138
M38549	*d*	34.925	B138
39236	*d*	60.000	B150
39250	*d*	63.500	B150
39412	*D*	104.775	B150
39520	*D*	112.712	B150,B152
39521	*D*	112.712	B152
39585	*d*	63.500	B150
39590	*d*	66.675	B152
41100	*d*	25.400	B134
41125	*d*	28.575	B134
41126	*d*	28.575	B134
41286	*D*	72.626	B134
42350	*d*	88.900	B158
42362	*d*	92.075	B160
42368	*d*	93.662	B160
42375	*d*	95.250	B160
42376	*d*	95.250	B160
42381	*d*	96.838	B160
42584	*D*	148.430	B160
42587	*D*	149.225	B158,B160
42620	*D*	127.000	B154,B156
42687	*d*	76.200	B154
42688	*d*	76.200	B154

호칭번호 CONE, CCUP	호칭치수(mm) *d*:CONE 내경 *D*:CUP 외경		베어링記載페이지
42690	*d*	77.788	B156
43118	*d*	30.162	B136
43131	*d*	33.338	B138
43300	*D*	76.200	B136
43312	*D*	79.375	B138
44143	*d*	36.512	B140
44150	*d*	38.100	B140
44157	*d*	40.000	B140
44162	*d*	41.275	B142
44348	*D*	88.501	B140,B142
L44610	*D*	50.292	B134
L44640	*d*	23.812	B134
L44643	*d*	25.400	B134
L44649	*d*	26.988	B134
45220	*D*	104.775	B148
45221	*D*	104.775	B148
45289	*d*	57.150	B148
L45410	*D*	50.292	B136
L45449	*d*	29.000	B136
46143	*d*	36.512	B140
46162	*d*	41.275	B142
46176	*d*	44.450	B142
46368	*D*	93.662	B140,B142
46720	*D*	225.425	B164
46780	*d*	158.750	B164
47420	*D*	120.000	B152,B154
47487	*d*	69.850	B152
47490	*d*	71.438	B154
47620	*D*	133.350	B154,B156
47680	*d*	76.200	B154
47685	*d*	82.550	B156
47686	*d*	82.550	B156
47687	*d*	82.550	B156
47820	*D*	146.050	B160
47890	*d*	92.075	B160
47896	*d*	95.250	B160
48120	*D*	161.925	B162
48190	*d*	107.950	B162
48220	*D*	182.562	B164
48282	*d*	120.650	B164
48286	*d*	123.825	B164
48290	*d*	127.000	B164
48320	*D*	190.500	B164
48385	*d*	133.350	B164
48393	*d*	136.525	B164
LM48510	*D*	65.088	B138
LM48511	*D*	65.088	B138
LM48548	*d*	34.925	B138
48620	*D*	200.025	B164
48685	*d*	142.875	B164
49175	*d*	44.450	B142
49176	*d*	44.450	B142
49368	*D*	93.662	B142
49520	*D*	101.600	B146